SAKSHAM JOSHI

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EDUCATION

Master's in Computer Application, Graphic Era University, Dehradun. 2025 - 2027
Coursework: Operating System, Data Structures and Algorithms, Full Stack Web Development.

Bachelor's in Computer Application, Graphic Era Hill University, Bhimtal. 2021 - 2024
Coursework: C and C++ Programming, Agile Software Development, Database Management System.

EXPERIENCE

Software Engineer Intern at Arterns Technologies Pvt Ltd May 2025 - November, 2025

- Developed an employee verification API using Bun, Elysia.js, Redis, and PostgreSQL, capable of handling more than 20,000 requests per second.
- Developed a static file hosting service to quickly download images, audios, videos, documents, etc., using FastAPI, Python, and Ngrok, capable of handling 10,000 requests per second.
- Implemented the entire login system using Next.js and Redis, writing Lua scripts for Redis (deployed on Upstash) to power an OTP verification login flow for a client's system while ensuring rate limiting to prevent attacks.

PROJECTS

SRON Programming Language Sept 2023 - Mar 2026

- Developed a high-performance, **cross-platform programming language from scratch** which provides 90+ built-in functions, 101 keywords and 9 data types achieving extremely fast runtime speed.
- Engineered compiler and interpreter from scratch using C++ (GCC & Clang) providing features like automatic garbage collection, file manipulation, cross-platform binary bytecode, low memory usage and much more.
- Released 6 stable versions (latest: v2.3, Jan 2026) with cross-platform compatibility for Windows and Linux, ensuring 100% consistent behavior across operating systems.
- Developed a Vscode extension and wrote extensive technical documentation with 100+ sample programs and tutorials, making the language easy to use for developers of all skill levels.
- **Link:** Source Code — Documentation — Download

OPEN SOURCE CONTRIBUTION

MasterPortfolio: Implemented a responsive PDF rendering page with React-PDF for direct viewing and downloading resume and also fixed the responsiveness of navbar and experience cards. View all PRs.

CosmoPortfolio: Fixed responsiveness of the navbar and other components. Optimized SEO, laggy UI and slow response by implementing lazy loading, conditional rendering, and splitting the site into multiple pages. View all PRs.

TECHNICAL SKILLS

Languages	C, C++, Python, Typescript
Web Technologies	Bun, Elysia.js, FastAPI, Flask, HTML, CSS, Next.js, React.js, Tailwind CSS
Cloud Technologies	Supabase, Render, Netlify, Ngrok, Upstash, GitHub Pages, Vercel, Hostinger
Databases	PostgreSQL, Redis, MySQL, Drizzle ORM
Other Tools	Linux, Bash, GCC, Clang, GitHub, Postman, Npm, Pnpm